

01_SYSTEM_ARCHITECTURE

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EVIDENCE PROVENANCE: [SYSTEM ARCHITECTURE]

Authoritative system boundary definitions, single state store hierarchy, and non-negotiable safety rules.

1. Architectural Hierarchy & Invariants

FOG-ORCHESTRATOR 2.0 enforces three core rules: (1) One authoritative Digital Twin state store (twin_state_store.py)--clients never invent state; (2) Inviolable Tier-1 local safety governor where dispatch cannot exceed safe speed; (3) Driver-in-the-loop assistance where the operator receives advisory warnings and remains in control.

2. Subsystem Definitions

- **Physical Prototype:** Dual ESP32 microcontrollers with optical tachometer, MPU6050 IMU, L298N/TB6612FNG H-bridges, SX1278 LoRa, and TWAI CAN.
- **Ingestion Gateway:** Rejects malformed packets, NaN/Inf, stale timestamps (>250ms), and out-of-order sequence counters.
- **Authoritative Digital Twin:** Maintains real spatial coordinates, ramp grades, road friction, and vehicle headway relationships.
- **Safety & Physics Solver:** Calculates continuous safe speed $v_{safe} = \min(v_{stop}, v_{retarder}, v_{traction}, v_{curve}, v_{mine})$.
- **Control Room HMI (5173):** Supervisory React 19 dashboard showing 3D corridor, fleet occupancy, and fog fronts.
- **In-Cab Operator Cockpits:** High-contrast driver consoles (Ports 3001/3002) with live speed, safe speed, and warnings.